

*MA.6.NSO.3.2***Benchmark**

**MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.**

Benchmark Clarifications:

*Clarification 1:* Instruction includes using the distributive property to generate equivalent expressions.

**Connecting Benchmarks/Horizontal Alignment**

- MA.6.AR.1.4
- MA.6.NSO.3.1, MA.6.NSO.3.4
- MA.7.AR.1.2

**Terms from the K-12 Glossary**

- Area Model
- Composite Number
- Factor
- Greatest Common Factor (GCF)
- Prime Number
- Rectangular Array

**Vertical Alignment****Previous Benchmarks**

- MA.4.AR.3.1
- MA.5.NSO.2.1, MA.5.NSO.2.2

**Next Benchmarks**

- MA.7.NSO.1.1
- MA.8.AR.1.1, MA.8.AR.1.3

**Purpose and Instructional Strategies**

In previous courses, students determined factor pairs for whole numbers from 0 to 144 and determined if numbers are prime, composite, or neither. Students also multiplied and divided using products and divisors greater than 144. In grade 6 accelerated, students use their understanding of factor pairs to find common factor pairs between addends to generate equivalent expressions, including the use of the distributive property. In future courses, students generate equivalent expressions including the use of integer exponents as well as rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

- This benchmark builds upon student understanding of the distributive property and supports the decomposition of numbers in earlier grades and extends to future learning in algebraic reasoning in future grade levels.
- Students should not use the multiplication symbol “ $\times$ ” when rewriting composite numbers as a common factor multiplied by the sum of two whole numbers. In grade 3, students have seen the multiplication symbol “ $\times$ ” when using the distributive property. Students should move away from this practice in grade 6 accelerated to writing the common factor directly outside the parentheses with an understanding that a number directly next to parentheses means you multiply (*MTR.3.1*).
  - For example,  $24 + 12 = 6(4 + 2)$ .
- Instruction includes the use of manipulatives, models, drawings and equations to rewrite the sum.

- If two numbers have multiple common factors, there will be multiple equivalent common factors multiplied by the sum of two whole number possibilities (*MTR.2.1, MTR.5.1*).
  - For example, the equations below are equivalent:  
 $42 + 96 = 2(21 + 48)$     $42 + 96 = 3(14 + 32)$     $42 + 96 = 6(7 + 16)$
- Provide opportunities for students to identify the common factor and the factors to sum in the parenthesis. Given the common factor multiplied by the sum of two whole numbers, determine the two composite whole numbers to be summed.
- For this benchmark, all terms are numerical, not algebraic.
- Students should use their basic multiplication facts to do this benchmark and think through number relationships and patterns (*MTR.5.1*).

### Common Misconceptions or Errors

- When working with the distributive property, some students may incorrectly multiply only the common factor by only one of the terms inside the parenthesis. Students need to make sure to multiply the common factor by both terms in the parentheses.
- Students may incorrectly think that they always must factor out the greatest common factor.
- Students must read the task or question carefully to determine if any common factor is allowable or if it must be the greatest common factor.

### Strategies to Support Tiered Instruction

- Instruction includes the use of area models to visually represent an application of the distributive property.
  - Teacher provides review of prior knowledge of the distributive property using an area model. For example, the teacher can begin with the expression  $5(2 + 3)$  and the area model below.

|   |   |   |
|---|---|---|
|   | 2 | 3 |
| 5 | ? | ? |

Then the teacher can co-creates the area model to demonstrate the distributive property.

|   |                      |    |
|---|----------------------|----|
|   | $5(2 + 3) = 10 + 15$ |    |
|   | 2                    | 3  |
| 5 | 10                   | 15 |

- Teacher models using an area model to determine a common factor. For example, the teacher can begin with the expression  $9 + 21$  and the area model below.

|   |   |    |
|---|---|----|
|   | ? | ?  |
| ? | 9 | 21 |

Then the teacher can co-creates the area model to determine a common factor.

|   |   |    |
|---|---|----|
|   |   |    |
|   | 3 | 7  |
| 3 | 9 | 21 |

- Instruction includes providing students with incomplete area models they must complete and use to write equivalent expressions.

### Instructional Tasks

#### *Instructional Task 1 (MTR.3.1, MTR.4.1)*

Rewrite  $121 + 66$  in the form  $a(b + c)$  where  $a$  is one of the common factors of 121 and 66 and  $b$  and  $c$  are whole numbers. Does your expression have the same value as  $121 + 66$ ? Justify your thinking.

### Instructional Items

#### *Instructional Item 1*

Rewrite the following numerical expression in an equivalent form using the distributive property:  $24 + 36$ .

#### *Instructional Item 2*

Rewrite  $2(4 + 8)$  as a numerical expression.

*\*The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*